

APPASA Strategy Document

Stage 5 – Share: Data Visualization, Storytelling & Business Reporting



APPASA: Share Stage

Introduction

I will use this **APPASA strategy document** to record my decisions and reflections as I work through **Stage 5: Share** of this project. This document will serve as a guide, helping me consider my responses and reflections throughout this stage of the data analysis process.

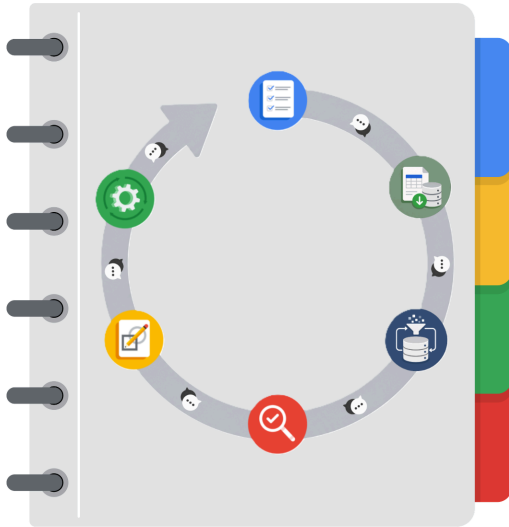
Portfolio Project Recap

Many of the goals I accomplished throughout the individual course portfolio projects are integrated into this end-to-end data analysis capstone project. These include:

- Creating a clear and structured project proposal.
- Demonstrating my understanding of SQL and relational database concepts.
- Using SQL to ingest, explore, clean, transform, validate, and organize data through structured queries and database operations.
- Organizing and analyzing a large-scale dataset to uncover meaningful insights and communicate the underlying business story.
- Developing a series of SQL scripts for ELT, feature engineering, data validation, and analytical querying, alongside a DuckDB-powered Jupyter Notebook documenting the complete APPASA methodology and reproducing the analytical workflow through SQL queries and visualizations.
- Computing descriptive statistics, performing exploratory and business-focused analyses, and creating analytical visualizations to support data interpretation and executive decision-making.
- Communicating findings through an executive summary, an interactive Power BI report, strategic business recommendations, and implementation-focused action plans for stakeholders.
- Translating analytical findings into evidence-based business recommendations, implementation strategies, success metrics, and future analytical opportunities through the final Act stage of the APPASA framework.

This capstone brings together the knowledge and technical skills I developed throughout the program, demonstrating my ability to execute a complete end-to-end data analysis project, from business problem definition to actionable business recommendations.

THE APPASA WORKFLOW



I will demonstrate to the company's marketing analytics team how I would apply the APPASA workflow to the upcoming Cyclistic Bike Share project. For each question presented in this APPASA strategy document, I will provide a structured and thoughtful response aligned with the corresponding stage of the APPASA framework: Ask, Prepare, Process, Analyze, Share and Act. This approach ensures clarity in my methodology, highlights my strategic planning skills, and illustrates a well-organized path from problem understanding to actionable insights.

Project Tasks

The following questions have been identified as essential to the Member vs Casual Rider Behavior Analysis Project. These questions are primarily situated within the Share stage of the APPASA workflow, emphasizing data visualization, dashboard design, business storytelling, and communicating analytical findings through clear, evidence-based reporting. I will address each question by drawing on relevant materials from the SQL scripts, project notebook, the APPASA framework, dashboard, and best practices in data visualization and business communication.





Data Project Questions & Considerations

Visualization Planning

- What communication objectives guided the dashboard design?

The primary communication objective was to transform the analytical findings into a clear, executive-friendly dashboard that effectively answers the project's business question: **how annual members and casual riders use Cyclistic bikes differently**. The dashboard was designed to present key performance indicators, rider behavior patterns, station usage, and actionable business recommendations in a logical sequence that supports informed decision-making.

Rather than displaying all analyses on a single page, the dashboard was structured to progressively guide stakeholders from a high-level overview to detailed behavioral insights, geographic patterns, and strategic recommendations. This approach enables executives to quickly understand the overall performance while providing sufficient detail to explore the underlying factors driving rider differences.

Interactive slicers for bike type, month, day of week, and rider type were incorporated to allow stakeholders to explore the data dynamically without overwhelming the report. Consistent formatting, colour schemes, and chart selection were also used to improve readability, highlight key comparisons, and ensure that the dashboard communicates insights efficiently to a non-technical business audience.

- How was the dashboard structured to communicate the analysis?

The dashboard was structured as a **four-page analytical narrative** that progressively guides stakeholders from a high-level performance overview to detailed behavioural insights, location-based analysis, and actionable business recommendations. This sequential design ensures that each page builds upon the previous one, allowing decision-makers to understand not only *what* differences exist between annual members and casual riders, but also *why* those differences matter from a business perspective.

The first page, **Executive Overview**, presents key performance indicators and high-level trends to establish the overall ridership landscape. The second page, **Rider Behavior Analysis**, explores behavioural differences through ride patterns, bike preferences, and ride duration distributions. The third page, **Ride Location Analysis**, identifies the most frequently used stations and compares location preferences between rider groups to highlight geographic usage patterns. Finally, the **Strategic Recommendations** page synthesizes the analytical findings into practical, data-driven business recommendations and identifies opportunities to convert casual riders into annual members.

This structured flow enables stakeholders to move naturally from understanding the current state of the business to interpreting rider behaviour and ultimately evaluating evidence-based recommendations that support Cyclistic's membership growth strategy.



- Why was a multi-page dashboard selected instead of a single report page?

A multi-page dashboard was selected to organize the analysis into logical, focused sections while maintaining clarity and readability for stakeholders. Attempting to present all key performance indicators, rider behavior analyses, station insights, and business recommendations on a single page would have resulted in a cluttered report, making it more difficult for users to identify important findings and interpret the analysis effectively.

Each dashboard page was designed to serve a distinct analytical purpose. The **Executive Overview** provides a high-level summary of Cyclistic's overall performance, the **Rider Behavior Analysis** examines behavioural differences between annual members and casual riders, the **Ride Location Analysis** explores station-level usage patterns, and the **Strategic Recommendations** page translates the analytical findings into actionable business initiatives. This separation allows each topic to be explored in sufficient detail without overwhelming the audience.

Organizing the dashboard into multiple pages also creates a natural storytelling flow, enabling stakeholders to progress from summary metrics to detailed analysis and ultimately to evidence-based recommendations. This structure improves the overall user experience, supports more effective decision-making, and aligns with dashboard design best practices by presenting information in manageable, logically connected sections.

- How were visualizations mapped to the project's business question?

The visualizations were intentionally selected to address different aspects of the project's primary business question: **how annual members and casual riders use Cyclistic bikes differently**. Rather than presenting unrelated charts, each visualization was designed to examine a specific dimension of rider behavior and collectively provide a comprehensive understanding of the differences between the two rider groups.

The **Executive Overview** page uses KPI cards and trend charts to establish high-level differences in ride volume, average ride length, monthly ridership trends, hourly activity, and weekly usage patterns. These visuals provide an overall comparison of rider behaviour before exploring more detailed analyses.

The **Rider Behavior Analysis** page focuses on behavioural characteristics by examining ride activity throughout the day, bike type preferences, and ride duration distributions. Where the objective was to compare behavioural composition independent of rider volume, percentage-based visualizations were used. Conversely, when analysing actual demand patterns, raw ride counts were presented to preserve the magnitude of rider activity.

The **Ride Location Analysis** page investigates spatial usage patterns by identifying the busiest start and end stations, comparing station usage between rider groups, and highlighting stations associated with the longest average ride durations. These visualizations reveal where rider behaviour differs geographically and support operational and marketing insights.

Finally, the **Strategic Recommendations** page synthesizes the findings from the previous analyses into evidence-based business recommendations. Rather than introducing additional charts, it communicates how the analytical insights can be applied to develop targeted strategies for converting casual riders into annual members.



By assigning each visualization to a specific analytical objective, the dashboard maintains a clear narrative flow and ensures that every visual directly contributes to answering the project's business question.

- Why was Power BI selected as the visualization platform?

Power BI was selected as the visualization platform because it provides a comprehensive business intelligence environment for transforming processed data into interactive, professional dashboards that support data-driven decision-making. Its robust data modeling capabilities, interactive visualizations, and user-friendly interface make it well suited for communicating analytical findings to both technical and non-technical stakeholders.

The processed dataset generated during the Process stage was imported into Power BI, where relationships, calculated measures, and a dedicated date table were created to support dynamic analysis. Interactive features such as slicers and cross-filtering enable users to explore rider behaviour by rider type, bike type, month, and day of the week without modifying the underlying dataset, making the dashboard flexible and easy to navigate.

Power BI also offers a wide range of customizable visualizations that effectively communicate trends, comparisons, distributions, and geographic usage patterns. These capabilities allowed the dashboard to present complex analytical findings in a clear and accessible manner while maintaining a consistent design across all report pages.

Overall, Power BI was chosen because it combines data modeling, interactive exploration, and high-quality visualization within a single platform, enabling the analytical results to be translated into meaningful business insights and actionable recommendations for Cyclistic's stakeholders.

Visualization Design & Dashboard Architecture

- Why were these visualization types selected?

The visualization types were selected based on the nature of the data and the analytical objective of each section, ensuring that every visual communicated insights clearly and supported meaningful comparisons between annual members and casual riders. Rather than choosing charts for visual appeal, each visualization was matched to the type of information being presented and the business question it was intended to answer.

KPI cards were used to highlight key metrics such as total rides, average ride duration, and rider counts, allowing stakeholders to quickly assess overall performance. **Line charts** were selected to display monthly, weekly, and hourly trends because they effectively illustrate changes in rider activity over time. **Column and bar charts** were used to compare categorical variables, including bike type preferences and station usage, making it easy to identify differences in rider behaviour across categories. **Percentage-based stacked bar charts** were chosen where the objective was to compare the relative distribution of rider preferences rather



than absolute ride volumes. **Matrix heatmaps** were incorporated to visualize rider activity across combinations of days and hours, enabling patterns of peak usage to be identified efficiently.

Each visualization was designed to answer a specific analytical question while complementing the others within the dashboard. Collectively, the selected visuals simplify complex datasets, emphasize the differences between rider groups, and communicate findings in a format that supports informed business decision-making.

- How do the chosen visualizations support comparisons between annual members and casual riders?

The dashboard visualizations were specifically designed to facilitate direct comparisons between annual members and casual riders across multiple dimensions of rider behaviour. Consistent colour coding, standardized chart formats, and side-by-side comparisons were used throughout the dashboard to help stakeholders quickly identify similarities, differences, and trends between the two rider groups.

Charts comparing ride volume, average ride duration, hourly activity, weekly riding patterns, bike type preferences, and station usage display both rider groups within the same visual, enabling meaningful comparisons without requiring users to switch between separate charts. This design makes differences in riding behaviour immediately apparent while maintaining consistency across the dashboard.

Where comparisons focused on rider composition or preference, **percentage-based visualizations** were used to normalize the data and eliminate the influence of differing ride volumes between the two groups. Conversely, when the objective was to evaluate operational demand or overall usage, **absolute ride counts** were presented to accurately reflect the scale of rider activity.

Interactive slicers further enhance comparisons by allowing stakeholders to filter both rider groups simultaneously across dimensions such as month, bike type, and day of the week. This enables users to investigate how behavioural differences change under different conditions while preserving a consistent comparison framework.

By combining consistent visual design, appropriate chart selection, normalized metrics where necessary, and interactive filtering, the dashboard provides a comprehensive and intuitive comparison of annual member and casual rider behaviour, directly supporting Cyclistic's objective of identifying opportunities to increase annual memberships.

- Which visualizations communicate proportions rather than raw counts, and why?

Percentage-based visualizations were used whenever the objective was to compare the relative behaviour and preferences of annual members and casual riders rather than the total number of rides. Since annual members generated substantially more trips than casual riders, relying solely on raw ride counts could have obscured meaningful behavioural differences by emphasizing differences in rider volume instead of rider characteristics.

In the dashboard, 100% stacked bar charts were used to compare variables such as bike type preferences and ride duration distributions. These visualizations normalize the data by expressing each category as a percentage of the total rides within each rider group, allowing stakeholders to compare behavioural patterns independently of group size.

Using proportions makes it easier to identify differences in preferences. For example, it highlights whether casual riders have a greater tendency toward longer ride durations or different bike type choices relative to annual members, regardless of the fact that members complete more rides overall. This provides a fair and meaningful comparison of rider behaviour.

In contrast, visualizations intended to represent operational demand, such as monthly ridership trends, station usage, and total ride volume, were displayed using raw counts because the actual magnitude of rider activity is essential for understanding usage patterns and supporting business decisions. By combining proportional and count-based visualizations where appropriate, the dashboard communicates both relative behavioural differences and overall rider demand in a balanced and informative manner.

- How were KPI metrics selected?

The KPI metrics were selected to provide an immediate summary of the performance indicators most relevant to the project's business question: **how annual members and casual riders differ in their usage of Cyclistic bikes**. Each KPI was chosen because it represents a fundamental measure of rider behaviour and serves as a starting point for the more detailed analyses presented throughout the dashboard.

The selected KPIs include **Total Rides**, **Average Ride Duration**, **Annual Member Rides**, **Casual Rider Rides**, **Average Ride Duration for Annual Members**, and **Average Ride Duration for Casual Riders**. Together, these metrics provide a concise overview of overall ridership, the distribution of riders between membership types, and differences in ride duration, enabling stakeholders to quickly understand the scale and nature of rider behaviour.

The KPIs were also selected to complement the supporting visualizations rather than replace them. While the KPI cards present high-level summary metrics, the accompanying charts explain the underlying trends, behavioural patterns, and geographic differences that contribute to those values. This combination allows users to move naturally from summary statistics to detailed analysis without losing context.

By focusing on metrics that directly align with the project's objectives, the KPI section provides an effective executive summary and establishes a clear foundation for interpreting the analytical findings presented in the rest of the dashboard.

- How were filters and slicers incorporated into the dashboard?

Interactive filters and slicers were incorporated throughout the dashboard to enable stakeholders to explore the data dynamically while maintaining a consistent analytical framework. Rather than creating multiple static reports, slicers allow users to customize the displayed information based on specific categories and immediately observe how rider behaviour changes across different conditions.

The dashboard includes slicers for **rider type**, **bike type**, **month**, and **day of the week**, allowing users to filter visualizations according to their analytical needs. These slicers are synchronized across the relevant report pages so that selected filters are applied consistently to all associated visualizations, ensuring meaningful comparisons and reducing the need for repetitive user interactions.

Power BI's cross-filtering functionality further enhances the dashboard by allowing interactions between visualizations. Selecting a category or data point within one chart automatically updates the related charts on the page, enabling users to investigate relationships between different aspects of rider behaviour without modifying the underlying dataset.

By incorporating interactive filters and slicers, the dashboard becomes more flexible and user-centric, allowing stakeholders to explore seasonal patterns, bike preferences, and rider characteristics from multiple perspectives while preserving a clear and consistent presentation of the analytical findings.

- Why were certain insights displayed as KPI cards instead of charts?

KPI cards were used for insights that represent **high-level summary metrics** rather than trends, distributions, or comparisons. Their purpose is to provide stakeholders with an immediate overview of the dashboard's most important performance indicators before they examine the detailed visualizations.

Metrics such as **Total Rides**, **Average Ride Duration**, **Annual Member Rides**, **Casual Rider Rides**, **Average Ride Duration for Annual Members**, and **Average Ride Duration for Casual Riders** are single aggregated values that can be understood quickly without graphical representation. Displaying these metrics as KPI cards reduces visual complexity and enables users to identify the overall scale and composition of rider activity at a glance.

Charts were reserved for analyses involving relationships, comparisons, and trends across multiple categories or time periods, where graphical representation provides greater analytical value. For example, line charts illustrate changes in ridership over time, bar charts compare bike type preferences and station usage, and heatmaps reveal patterns of rider activity across days and hours. These visualizations communicate information that cannot be effectively represented by a single numeric value.

Using KPI cards alongside charts creates a clear information hierarchy within the dashboard. The KPI cards provide a concise executive summary, while the accompanying visualizations explain the underlying patterns and factors that contribute to those metrics. This combination improves readability, supports faster decision-making, and allows stakeholders to transition naturally from high-level performance indicators to detailed behavioural analysis.

- Why was the dashboard organized into four sequential pages, and how does this structure guide users through the analysis?

The dashboard was organized into four sequential pages to present the analysis as a logical business narrative rather than a collection of independent visualizations. Each page has a distinct analytical purpose and builds upon the insights from the previous page, enabling stakeholders to progressively understand the data,

identify behavioural patterns, and arrive at evidence-based business recommendations. This structured approach improves readability, reduces information overload, and supports informed decision-making.

The dashboard begins with the **Executive Overview**, which provides a concise summary of key performance indicators, overall ridership trends, and high-level comparisons between annual members and casual riders. This page establishes the analytical context and allows stakeholders to quickly understand the scale and composition of the dataset before exploring more detailed findings.

The second page, **Rider Behavior Analysis**, investigates how the two rider groups differ in their riding habits. Visualizations examine ride duration, hourly and weekly usage patterns, and bike type preferences, revealing behavioural differences that directly address the project's business question.

The third page, **Ride Location Analysis**, extends the analysis by examining spatial usage patterns, including the most frequently used stations and location-based rider behaviour. This page provides geographic context that helps explain where demand is concentrated and how rider groups utilize the bike-share network differently.

The final page, **Strategic Recommendations**, synthesizes the analytical findings into actionable business recommendations. Rather than introducing new analyses, this page connects the evidence presented throughout the dashboard to practical strategies aimed at increasing annual memberships, ensuring that stakeholders can clearly understand how the analysis supports the proposed recommendations.

By following this sequential structure, moving from the executive summary to behavioural analysis, then to geographic insights, and finally to strategic recommendations, the dashboard guides users through a coherent analytical journey. Each page answers a specific set of questions while naturally leading to the next stage of analysis, resulting in a clear, evidence-based narrative that effectively supports business decision-making.

- How do the dashboard pages work together to communicate the complete analytical story?

The four dashboard pages were designed to function as a connected analytical narrative, with each page contributing a distinct layer of insight while building upon the findings of the previous one. Rather than presenting isolated visualizations, the dashboard guides stakeholders through a structured journey from understanding the overall performance of the bike-share system to identifying behavioural patterns and ultimately arriving at data-driven business recommendations.

The **Executive Overview** establishes the foundation by summarizing the key performance indicators and presenting an overview of ridership trends. This page answers the question, "*What does the overall data look like?*" and provides the context needed for the subsequent analyses.

Building on this foundation, the **Rider Behavior Analysis** page examines *how* annual members and casual riders differ in their riding habits. By exploring ride duration, riding frequency, bike type preferences, and temporal usage patterns, this page identifies the behavioural differences that explain the high-level metrics presented in the Executive Overview.

The **Ride Location Analysis** page then adds a geographic perspective by showing *where* these behaviours occur. Analysing station usage and location-based ride patterns helps stakeholders understand how spatial demand varies between rider groups and reveals opportunities for targeted operational or marketing strategies.

Finally, the **Strategic Recommendations** page synthesizes the evidence gathered throughout the previous pages into actionable business recommendations. Instead of introducing new analyses, it links the behavioural and geographic insights directly to strategies for converting casual riders into annual members, ensuring that every recommendation is supported by the findings presented earlier in the dashboard.

Together, these four pages create a cohesive analytical story that progresses from **what happened**, to **how riders behave**, to **where those behaviours occur**, and ultimately to **what actions Cyclistic should take**. This logical sequence enables stakeholders to understand not only the analytical findings but also how those findings support informed, evidence-based business decisions.

Business Storytelling & Communication

- Who is the intended audience for the dashboard?

The dashboard was designed primarily for **Cyclistic's executive leadership, marketing team, and business decision-makers** who are responsible for developing strategies to increase annual memberships. As these stakeholders may have varying levels of technical expertise, the dashboard emphasizes clear visual communication, intuitive navigation, and concise presentation of key findings rather than complex statistical analysis.

For executives, the dashboard provides high-level KPI metrics and summary visualizations that support rapid assessment of overall rider performance and membership trends. For marketing and business teams, the detailed behavioural and geographic analyses offer actionable insights into how casual riders differ from annual members, helping identify opportunities for targeted campaigns, promotional offers, and customer engagement strategies.

The dashboard is also suitable for analysts and project stakeholders who require a deeper understanding of the underlying data. Interactive filters, detailed visualizations, and supporting comparisons allow users to explore rider behaviour across different dimensions while maintaining a consistent analytical framework.

By balancing executive-level summaries with detailed analytical insights, the dashboard effectively serves both strategic decision-makers and analytical users, ensuring that the findings are accessible, actionable, and aligned with Cyclistic's objective of converting more casual riders into annual members.

- How does the dashboard answer the original business question?

The dashboard was designed specifically to answer the project's central business question: **How do annual members and casual riders use Cyclistic bikes differently?** It accomplishes this by presenting a

comprehensive comparison of the two rider groups across multiple behavioural, temporal, and geographic dimensions, allowing stakeholders to identify the patterns that distinguish casual riders from annual members.

The **Executive Overview** establishes the overall context by summarizing key performance indicators and ridership trends, highlighting differences in rider composition and average ride duration. The **Rider Behavior Analysis** then explores how usage varies between the two groups by examining ride duration, hourly activity, weekday usage, and bike type preferences. These visualizations reveal that annual members primarily use the service for regular weekday commuting with shorter, more consistent trips, while casual riders tend to take longer recreational rides, particularly during weekends and leisure periods.

The **Ride Location Analysis** complements these findings by identifying where riders begin and end their journeys, revealing location-based differences in travel behaviour and station usage. These geographic insights provide additional context for understanding how and where each rider group interacts with the bike-share system.

Finally, the **Strategic Recommendations** page connects the analytical findings to Cyclistic's business objective by translating the observed behavioural differences into practical marketing and operational recommendations aimed at encouraging casual riders to become annual members.

By integrating executive summaries, behavioural comparisons, geographic analyses, and evidence-based recommendations into a single analytical workflow, the dashboard provides a complete and data-driven answer to the original business question while directly supporting Cyclistic's goal of increasing annual memberships.

- What story do the visualizations communicate?

The visualizations collectively communicate the story of **how annual members and casual riders differ in their usage of Cyclistic's bike-share service and how these behavioural differences can inform strategies to increase annual memberships**. Rather than presenting isolated charts, the dashboard arranges the visualizations to progressively build an evidence-based narrative that answers the project's business question and supports strategic decision-making.

The story begins by establishing the overall performance of the bike-share system through KPI cards and summary visualizations, providing stakeholders with a clear understanding of rider composition, ride volume, and average ride duration. It then transitions into a detailed comparison of rider behaviour, illustrating that annual members primarily use the service for consistent weekday commuting with shorter rides, while casual riders tend to take longer, recreational trips that are concentrated during weekends and leisure hours.

The narrative is further developed through geographic visualizations, which demonstrate where rider activity is concentrated and highlight differences in station usage between the two rider groups. These spatial insights complement the behavioural analysis by revealing how location influences rider patterns and identifying areas where marketing or operational initiatives could be most effective.

Finally, the dashboard concludes by connecting these findings to practical business recommendations. The visualizations collectively demonstrate that the behavioural differences between casual riders and annual members present clear opportunities for targeted membership campaigns, seasonal promotions, and



location-specific marketing strategies aimed at encouraging casual riders to transition to annual memberships.

By progressing from **overall performance**, to **behavioural differences**, to **geographic insights**, and ultimately to **actionable recommendations**, the dashboard communicates a complete analytical story that transforms raw data into meaningful business insights and directly supports Cyclistic's strategic objective of increasing annual memberships.

- How were analytical findings translated into business recommendations?

The analytical findings were translated into business recommendations by identifying consistent behavioural patterns within the data and linking those patterns to Cyclistic's objective of increasing annual memberships. Rather than presenting observations in isolation, the analysis focused on understanding how differences in rider behaviour could be leveraged to develop targeted marketing and operational strategies.

The dashboard revealed that annual members primarily use the bike-share service for regular weekday commuting, while casual riders are more likely to take longer recreational trips during weekends and leisure hours. These behavioural differences indicated that casual riders value flexibility and leisure-oriented experiences, whereas annual members benefit from frequent and routine usage. Based on these insights, recommendations were developed to encourage casual riders to recognize the long-term value of an annual membership.

Geographic analysis further supported the recommendations by identifying stations and locations with high concentrations of casual rider activity. These findings suggest opportunities to implement location-specific promotional campaigns, membership advertisements, and targeted outreach in areas where conversion potential is highest. Seasonal ridership trends and usage patterns also informed recommendations for offering promotional discounts and marketing campaigns during periods of increased casual rider activity.

Each recommendation presented in the dashboard is directly supported by evidence from the behavioural, temporal, and geographic analyses, ensuring that proposed actions are grounded in data rather than assumptions. By translating analytical insights into practical strategies, the dashboard bridges the gap between data analysis and business decision-making, providing Cyclistic with actionable recommendations to support its goal of converting more casual riders into annual members.

- How do the recommendations relate to the analysis?

The recommendations are directly derived from the analytical findings and represent the practical application of the patterns identified throughout the dashboard. Rather than being based on assumptions or general best practices, each recommendation is supported by evidence obtained from the behavioural, temporal, and geographic analyses conducted during the project.

The analysis showed that annual members primarily use Cyclistic bikes for consistent weekday commuting, while casual riders exhibit longer ride durations and higher usage during weekends and leisure periods. These behavioural differences indicate that casual riders interact with the service differently and therefore require

marketing strategies tailored to their specific usage patterns. Consequently, the recommendations focus on promoting the long-term value of annual memberships to riders who frequently use the service for recreational purposes.

Similarly, the geographic analysis identified stations and areas with high concentrations of casual rider activity, providing evidence for location-specific marketing initiatives and membership promotions. Seasonal and temporal usage trends also informed recommendations for launching targeted campaigns during periods when casual rider demand is highest, increasing the likelihood of successful membership conversions.

By ensuring that every recommendation can be traced back to one or more analytical findings, the dashboard maintains a clear connection between **data, insight, and action**. This evidence-based approach strengthens the credibility of the proposed strategies and demonstrates how the analysis supports Cyclistic's business objective of converting more casual riders into annual members.

- How does the dashboard support executive decision-making?

The dashboard supports executive decision-making by transforming complex analytical results into a clear, concise, and actionable business report. Rather than requiring stakeholders to interpret raw data or detailed statistical outputs, the dashboard presents the most important findings through intuitive visualizations, executive-level KPI summaries, and evidence-based recommendations that directly address Cyclistic's business objective of increasing annual memberships.

The **Executive Overview** provides an immediate snapshot of key performance indicators, enabling decision-makers to quickly assess overall rider activity, membership composition, and ride duration without reviewing the underlying dataset. Subsequent pages progressively explain the behavioural and geographic factors that contribute to these high-level metrics, allowing executives to understand not only *what* is happening but also *why* these patterns occur.

Interactive filters and synchronized visualizations further support decision-making by allowing stakeholders to explore specific rider segments, time periods, and bike types without compromising the overall analytical narrative. This flexibility enables executives to investigate areas of interest while maintaining consistent comparisons across the dashboard.

Most importantly, the dashboard concludes with a dedicated **Strategic Recommendations** page that translates analytical findings into practical business actions. By linking every recommendation directly to evidence presented in the preceding analyses, the dashboard ensures that strategic decisions are supported by data rather than assumptions. This evidence-based approach helps executives confidently identify opportunities for targeted marketing campaigns, membership promotions, and resource allocation, ultimately supporting Cyclistic's objective of converting more casual riders into annual members.



Accessibility & Presentation

- How was accessibility considered during dashboard design?

Accessibility was an important consideration throughout the dashboard design to ensure that the analytical findings could be easily understood by users with varying levels of technical expertise. The dashboard emphasizes clarity, consistency, and intuitive navigation so that executives, marketing teams, and analysts can quickly interpret the information and make informed decisions.

A consistent colour palette, typography, and formatting were applied across all four dashboard pages to create a cohesive user experience and reduce visual distractions. Charts were selected based on their ability to communicate information clearly, while descriptive titles, axis labels, legends, and data labels were included where appropriate to improve interpretation. Consistent colour coding was also used to represent annual members and casual riders throughout the dashboard, allowing users to identify rider groups immediately without repeatedly consulting legends.

The dashboard follows a logical reading order, progressing from executive-level summaries to detailed behavioural analysis, geographic insights, and strategic recommendations. This structured layout reduces cognitive load and enables stakeholders to understand the analytical story step by step. Interactive slicers were placed in consistent locations and synchronized across pages, ensuring that users could filter the data without disrupting the overall analytical workflow.

Visual clutter was minimized by avoiding unnecessary decorative elements and limiting each page to a focused set of related visualizations. High-contrast text, adequate spacing, and appropriately sized fonts further enhance readability across different screen sizes and presentation settings.

By prioritizing simplicity, consistency, and intuitive navigation, the dashboard ensures that its insights are accessible to both technical and non-technical stakeholders, enabling a wider audience to confidently interpret the findings and support data-driven business decisions.

- How were colours used to improve readability?

Colours were used strategically throughout the dashboard to improve readability, reinforce visual consistency, and help users quickly distinguish between key categories without overwhelming the interface. Rather than using numerous decorative colours, the dashboard follows a restrained and consistent colour scheme that supports effective data interpretation.

A consistent colour palette was applied across all four dashboard pages, with the same colours representing **annual members** and **casual riders** in every visualization. This consistency allows stakeholders to recognize rider groups immediately, reducing the need to repeatedly reference chart legends and making comparisons more intuitive. Neutral background colours and subtle accent colours were used to maintain visual balance while ensuring that the data remained the primary focus.

Colour was also used to establish visual hierarchy. KPI cards, chart titles, and important metrics were highlighted using accent colours to draw attention to key insights, while supporting information was displayed using more neutral tones. This helps users quickly identify the most important information without becoming distracted by less critical details.

High colour contrast between text, backgrounds, and chart elements was maintained to enhance readability and improve accessibility for a wide range of users. The limited use of colours also reduces visual clutter, allowing stakeholders to focus on patterns, trends, and comparisons rather than decorative design elements.

By applying colours consistently and purposefully, the dashboard improves readability, strengthens comparisons between rider groups, and creates a clear, professional interface that effectively communicates analytical insights to both technical and non-technical audiences.

- How were titles, subtitles, and labels designed?

Titles, subtitles, and labels were designed to enhance clarity, provide context, and guide users through the analytical narrative without requiring additional explanation. Each dashboard page includes a descriptive title that clearly communicates its purpose, while concise subtitles provide context about the analyses being presented and their relevance to Cyclistic's business objective.

Chart titles were written using clear, business-oriented language that describes the insight being presented rather than simply naming the visualization. This allows stakeholders to immediately understand the focus of each chart without needing to interpret technical terminology. Where appropriate, subtitles were used to provide additional context, explain the scope of the analysis, or indicate the dimensions being compared.

Axis labels, legends, and data labels were kept concise and consistent throughout the dashboard to improve readability. Standardized terminology was used across all visualizations to ensure that the same metrics, rider categories, and time periods were described consistently, reducing potential confusion. Labels were formatted using readable font sizes and positioned to avoid overlapping with chart elements, ensuring that important information remained visible and easy to interpret.

The naming convention for dashboard pages also follows the logical progression of the analytical story, moving from **Executive Overview** to **Rider Behavior Analysis**, **Ride Location Analysis**, and finally **Strategic Recommendations**. This consistent structure helps users navigate the dashboard intuitively while reinforcing the flow from summary metrics to business recommendations.

By using clear, descriptive, and consistently formatted titles, subtitles, and labels, the dashboard improves usability, minimizes ambiguity, and enables both technical and non-technical stakeholders to quickly understand and interpret the analytical findings.

- What design choices reduce cognitive load?

Several design choices were incorporated to reduce cognitive load and help users interpret the dashboard efficiently. The dashboard follows a clear and consistent layout that presents information in a logical



sequence, allowing stakeholders to focus on understanding the insights rather than navigating the interface. By dividing the analysis into four dedicated pages, **Executive Overview**, **Rider Behavior Analysis**, **Ride Location Analysis**, and **Strategic Recommendations**, the dashboard avoids overwhelming users with excessive information on a single screen.

Each page contains only a focused set of related visualizations, ensuring that users can concentrate on one aspect of the analysis at a time. Consistent placement of KPI cards, charts, slicers, and navigation elements further reduces the effort required to locate information, while synchronized filters allow users to explore different perspectives without learning new interactions on each page.

Visual consistency also contributes to reducing cognitive load. The dashboard uses standardized colours, typography, chart styles, and terminology across all visualizations, enabling users to recognize recurring patterns and comparisons quickly. Clear titles, concise labels, and descriptive legends provide sufficient context without adding unnecessary detail, while unnecessary decorative elements were intentionally avoided to keep attention on the data.

An intuitive visual hierarchy ensures that the most important information is seen first. Executive-level KPI cards are positioned prominently, followed by supporting charts that explain the underlying trends and relationships. This progression allows stakeholders to move naturally from summary metrics to detailed analysis and finally to strategic recommendations.

By emphasizing simplicity, consistency, and a structured analytical flow, the dashboard minimizes cognitive effort and enables both technical and non-technical stakeholders to interpret the findings quickly and make informed, data-driven decisions.

- Why was consistent formatting important?

Consistent formatting was essential to create a professional, easy-to-navigate dashboard that enables stakeholders to interpret information quickly and accurately. By applying the same design standards across all four dashboard pages, users can focus on understanding the analytical findings rather than adapting to different layouts or visual styles.

Uniform formatting was maintained through the consistent use of colours, typography, chart styles, KPI card designs, spacing, and page layouts. The same colour scheme was used to represent annual members and casual riders throughout the dashboard, while standardized fonts, titles, and labels ensured that similar information was presented in a familiar way across every visualization. This consistency reduces confusion and makes comparisons between charts more intuitive.

Consistent formatting also reinforces the dashboard's analytical narrative. As users progress from the **Executive Overview** to **Rider Behavior Analysis**, **Ride Location Analysis**, and **Strategic Recommendations**, they encounter a familiar visual structure that supports a smooth and logical flow of information. This predictable layout minimizes cognitive effort and allows stakeholders to concentrate on the insights being communicated.

From a business perspective, consistent formatting enhances the dashboard's credibility and professionalism. A well-organized and visually cohesive report improves stakeholder confidence in the analysis and facilitates effective communication during presentations and decision-making discussions.

By maintaining a consistent visual design throughout the dashboard, the analysis becomes easier to navigate, comparisons become more meaningful, and the overall user experience is significantly improved for both technical and non-technical audiences.

- Why was a dedicated Strategic Recommendations page included?

A dedicated **Strategic Recommendations** page was included to bridge the gap between data analysis and business decision-making. While the preceding dashboard pages focus on presenting and explaining analytical findings, the final page translates those insights into clear, actionable recommendations that directly support Cyclistic's objective of increasing annual memberships.

The dashboard progressively builds an analytical narrative by first summarizing overall performance, then examining rider behaviour and geographic usage patterns. The Strategic Recommendations page serves as the conclusion to this narrative by synthesizing the evidence from the previous analyses and demonstrating how the identified patterns can be used to inform marketing and operational strategies. This ensures that stakeholders are not only presented with data but are also provided with practical guidance on how to act upon the findings.

Separating the recommendations from the analytical visualizations also improves readability and prevents the dashboard from becoming cluttered. Decision-makers can review the supporting evidence throughout the earlier pages before referring to a concise summary of recommended actions, creating a logical progression from **data**, to **insight**, and finally to **business action**.

Each recommendation on this page is explicitly linked to findings from the behavioural, temporal, and geographic analyses, reinforcing that the proposed strategies are evidence-based rather than speculative. By presenting actionable insights in a dedicated section, the dashboard enables executives and marketing teams to quickly identify opportunities for targeted campaigns, membership promotions, and resource allocation.

Including a separate Strategic Recommendations page therefore enhances the dashboard's value as a business decision-support tool, ensuring that the analysis not only answers the original business question but also provides a clear path for achieving Cyclistic's strategic goal of converting more casual riders into annual members.

- How does the dashboard guide the audience from findings to recommendations?

The dashboard guides the audience from findings to recommendations by following a structured analytical narrative that progressively transforms raw data into actionable business insights. Rather than presenting visualizations as isolated analyses, each dashboard page builds upon the previous one, enabling stakeholders to understand not only the results but also their business implications.



The journey begins with the **Executive Overview**, which provides a concise summary of key performance indicators and overall ridership trends, establishing the context for the analysis. The **Rider Behavior Analysis** then explores the differences between annual members and casual riders by examining ride duration, hourly activity, weekday usage, and bike type preferences. These behavioural insights explain how the two rider groups interact with the bike-share service and identify the patterns that distinguish them.

The **Ride Location Analysis** extends the narrative by identifying where rider activity is concentrated and highlighting location-based differences in station usage. This additional layer of analysis provides geographic context, helping stakeholders understand where marketing efforts and operational improvements could have the greatest impact.

Finally, the **Strategic Recommendations** page synthesizes the evidence presented throughout the dashboard into clear, practical actions. Each recommendation is explicitly supported by findings from the behavioural, temporal, and geographic analyses, demonstrating how the observed patterns can be translated into targeted marketing campaigns, membership promotions, and operational strategies.

By progressing logically from **executive summary**, to **behavioural insights**, to **geographic analysis**, and ultimately to **business recommendations**, the dashboard creates a coherent flow from **data**, to **evidence**, to **decision-making**. This structured approach ensures that stakeholders can clearly understand how each recommendation is supported by the analysis, making the proposed actions both credible and actionable.